

Ames mutagenicity dataset (n = 13,560)

Drug-sourced: 6,416 (pos. 58.3%) | Industrial-sourced: 7,144 (pos. 3.0%)

Level 2a: Random-split CV

Stratified 5×5 CV

MCC = 0.670 ± 0.010

Level 2b: Scaffold CV

GroupKFold 5×5 CV

MCC = 0.624 ± 0.049

Level 1: Test set

80/20 held-out

MCC = 0.470

Level 3: Cross-source-domain leave-domain-out

Drug → Industrial

Train: 6,416 (pos. 58.3%) | Test: 7,144 (pos. 3.0%)

MCC = 0.234

Industrial → Drug

Train: 7,144 (pos. 3.0%) | Test: 6,416 (pos. 58.3%)

MCC = 0.122

Source-domain ($\Delta = 0.390$) >> Split method ($\Delta = 0.046$) $\approx$ Algorithm (range = 0.063)